

**Govt. Graduate College for Women Baghbanpura LHR**



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**Course Title :** Biological basis of Behavior

**Topic :** Biological research method of biopsychology

## **Biological Research Method of Biopsychology**

In 1861, Paul Broca, French neurologist, found that one of his patients who had lost the ability to speak, had damage in left frontal cortex. More patients with loss of speech showed damage in and around this area, which is now known as Broca's area.

Thus, the research conducted in this area concludes that brain damage causes effects like increase or decrease in hunger, changes in emotional responses, to mention just a few.

- Ablation method
- Historical method
- Psycho physiological Recording method
- Electrical Stimulation
- Chemical stimulation

### **Ablation Method**

Ablation means removal of a brain area, generally with a surgical knife. It is different from lesion. Both ablation and lesion methods are invasive and involve cutting a part of brain. Ablation method thus, involves the destruction of some tissue in the brain. This is followed by assessing the changes that occur in behavior due to the removal or destruction of the tissue. Any wound or injury thereby, is known as a lesion. But one has to be very careful as different parts of the brain are all connected to one another and work in conjunction with another. So if a lesion is created in one area, it may exert an influence on the neighbouring areas as well. Sometimes the animal also recovers from the brain damage either partially or the functions of the damaged part of the brain are covered up by another part of the brain. There are different ways to cause lesions. The main techniques discussed here are aspiration, radio frequency, knife cuts, cryogenic blockades and nerve poison.

### **Histological Methods**

With the advent of technology in the present times, there are many techniques and methods to study anatomy and functioning of human brain. For studying minute details of tissue (microscopic investigation), histological methods are used for the purpose. In this method, the sample of the brain that is under investigation, needs to be prepared so that it can be mounted on a slide and observed under a light microscope. The neural tissue is very fragile and under normal conditions it may just disintegrate. Hence, the tissue needs to be fixed and stained in order to be viewed on the slide. When it is taken out, the bacteria or moulds tend to decompose the neural tissue.

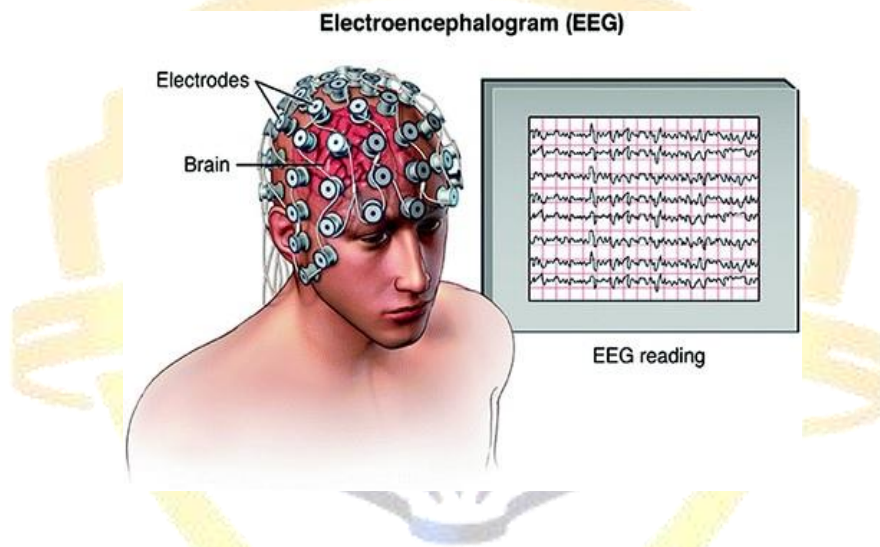
#### **Procedures in histological methods:-**

- **Fixation**
- **Freezing**
- **Embedding**
- **Staining**
- **Neuro imaging**

## Psycho physiological Recording Methods

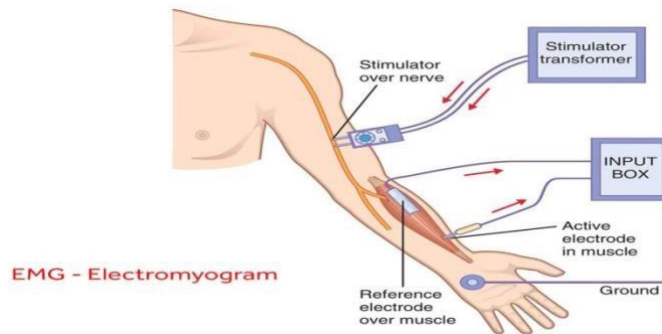
There are different psycho physiological methods that help to examine human participants in a noninvasive way. Such methods record from outside the skull (and other parts of the body) without inserting anything. The most widely used methods in this category are as follows:

**Electroencephalography:** The electrical activity of the brain is recorded using Electroencephalograph (EEG) machine and the technique is known as electroencephalography. Electrodes are placed on the scalp of the patient and recording is done on the computer. Earlier a rolling paper was used where the readings were displayed, now a computer printout is taken. The EEG signals help in the diagnosis of various disorders of the brain, mild head injury, cerebral pathology, epilepsy, and speed of processing information or how the memory functions with an increase in age.



**Magnetoencephalography:** Also referred as MEG, measures the changes in magnetic fields on the surface of the scalp, produced by changes in the basic patterns of the neural activity.

**Electromyography:** The muscle tension is recorded with the help of electromyogram (EMG). The electrodes are attached to the skin over the muscle that is to be investigated. When the muscle contracts producing muscle tension, it is recorded. This information helps us to know about muscle tone.



Picture Credit - Medical Dictionary

**Electro oculography** : The eye movements are recorded by using an electrooculogram (EOG). There is a slight difference in the electrical potential in the back and front portion of the eye ball. This difference in potentials is recorded with the help of electrodes that are placed near the eye



### Electrical Stimulation

The brain can be stimulated electrically to gain information about the working of its various neural systems. When the brain is stimulated, there is a change in the resting membrane potential and an action potential is initiated. This causes behavioral changes depending upon which part of the area has been stimulated. For stimulating the brain electrically, electrodes are used. Microelectrodes are made of fine metal wires through which the nerves are electrically stimulated. Macro electrodes are made of steel wires and are placed either on top of the scalp or inside the brain to record the functioning of tracts, which are large number of neurons together. The electrical recordings can be made over a long period of time, such as when the animal is recovering from any surgery. This is known as chronic recording. If the scientist is focusing to study a particular nerve pathway or where the information is going, then acute recording is done. These recordings of the electrical signals are then displayed through an oscilloscope or ink writing oscilloscope.

**Oscilloscope**: It helps to display the electrical recordings per unit time. It has a cathode ray tube on which the display is done. It is used to record the activity of one single neuron. When the action is detected, it is amplified and easily heard by researcher.

**Computer**: Helps to display, store & analyze the information that is collected by recording the electrical signals.

### **Chemical Stimulation:-**

There are various ways to chemically stimulate the brain and make lesions. Drugs can be given by injections or by inserting a tube into the stomach directly or into a vein or fatty tissue in the body. The chemical activity of the brain can be assessed by using various techniques. One is by using 2deoxyglucose technique (2-DG), 2-DG is a radioactive drug and when this drug is injected in the animal, the researcher waits till it is absorbed by the brain.

### **Neuroimaging :-**

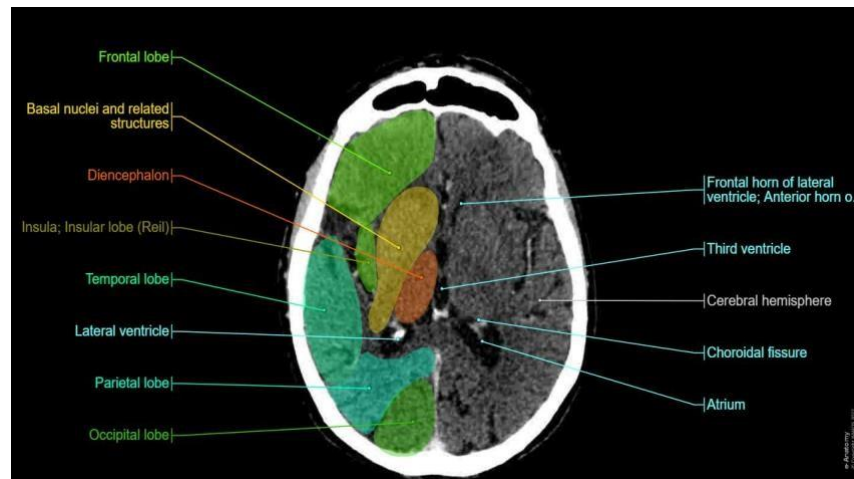
Brain imaging is known as neuro-imaging. Structural imaging methods are used to image the structure of the brain. These procedures help to diagnose any injury or disease in the brain as tumors. Functioning imaging helps to understand the function of the brain. It also helps in diagnosing diseases related to metabolism, or brain disorder or disease.

**X-ray technique:** Initially, this technique was used widely to explore the reason behind any neurological deficits. The conventional X-ray did not help much in observing the neural activity of a patient suffering from stroke or neurological disorders. It only provided information about the structures that absorbed the X- ray. The cerebral ventricular system and circulatory system within the brain is examined by using the contrast X-ray technique. In this technique, a substance is injected into one compartment of the body that absorbs X-rays either less than or more than the surrounding tissue. This creates a contrast during X-ray photography. Pneumoencephalography and cerebral angiography are two ways in which contrast X-rays are used. The former is a very old radiographic technique to examine the brain. Cerebral angiograms are most useful for localizing vascular damage and help to detect any tumor

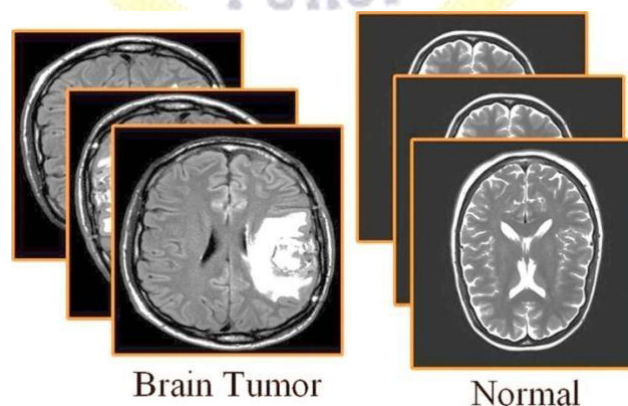


**Computerized Tomography (CT scan):** This is an X-ray technique used along with the computer that helps to view the structures of the body and the brain. There is a huge cylindrical ring in which the person lies down. The X-ray is passed from one side and the X-ray detector is placed on the other side of

the patient's head. Slowly the X-ray and the detector are rotated and the photographs are taken from all angles. All the photos are then collected by the computer and a CT scan of the brain is produced. A number of scans are taken and then they are combined to get a 3-D picture of the brain. Neurologists often use CT scans to diagnose blood clots, tumors, sclerosis or any internal injury, etc. and if surgery is to be carried out or not.

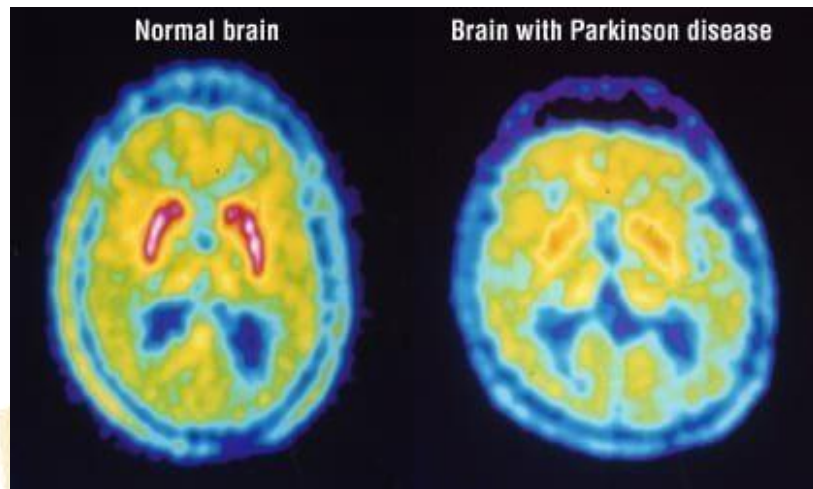


**Magnetic Resonance Imaging (MRI):** This technique is used when detailed picture of the patient's brain is to be investigated. This technique uses a strong magnetic field to gather images of the brain in frontal and horizontal planes. Unlike CT scan, it provides more detailed and clearer picture of the brain. This technique helps to know the blood oxygen flow to the different areas of the brain that are active. The resolutions of the images are high and changes are recorded as they occur in real time. It is safe as no injection is given to the patient.



**Positron Emission Tomography (PET):** PET scan helps to detect the changes in the metabolic activity taking place in the brain. It does not provide any structural information about the brain, rather it provides images of brain activity, that is, functional brain images. It is an invasive method where an

injection of radioactive 2-deoxyglucose (2-DG) is given to the patient. When it is absorbed by the patient, the patient is given a task to do and while doing the task, recordings are made. The images recorded show which part of the brain was actively involved in the task. PET scans use radioactive chemicals with a short shelf-life, made in cyclotron (a device to accelerate charged atomic or subatomic particles). It is an expensive technique hence, it is mostly available in research hospitals.



**Transcranial Magnetic Stimulation (TMS):** It is a technique that turns off a part of cortex by creating a magnetic field. It temporarily inactivates the neurons below the magnet. Researchers use TMS as a mapping technique. For instance, when TMS coil is placed on motor cortex and stimulating different regions along this cortical strip, muscle movements and twitches will be observed on the contralateral side of the body. From this, it can be observed how different body parts map onto different parts of the cortex and also how much cortex is devoted to any particular body part. The main disadvantage of TMS is that it is not very effective in reaching the subcortical structures